AI Compute and Educational Access: Living Research Notebook

Working hypotheses and a non-exhaustive candidate forecast

4 September 2026 | Working snapshot 0.1.0

This is an ongoing research notebook about where AI-assisted educational translation and adaptation could create useful access. It is not the final study, an exhaustive list of needs, or a completed funding-priority ranking.

The 100 entries are an initial forecast of candidates worth investigating. Their identifiers are not ranks. A language missing from the forecast, or not yet studied in depth, has not been judged less important. Missing evidence is unfinished research, not evidence of no need.

Every educational stage remains eligible: early learning and primary school, secondary school, vocational and practical learning, university, and postgraduate or research use. The same language may have several different educational opportunities.

The study concerns worldwide educational needs. Files, translations or previous spending in one local programme do not reduce a population's need or determine its place in the comparison.

GitHub is the continually evolving version. This dated Zenodo snapshot preserves the hypotheses and research-coverage checkpoint available on 4 September 2026. Later evidence may change findings and recommendations; the original forecast is retained so changes are visible.

Living version: https://kokunoyumeto.github.io/allocating-ai-translation-compute-for-educational-access/

How to read this notebook

Forecast

An expectation to test, not a conclusion or a list of the only eligible communities.

Package proposal

A specified educational resource to compare. Writing a proposal does not establish its effectiveness.

Not yet investigated

Work remains. It does not mean low need, low priority, or no useful resource could be made.

Research coverage

How far this investigation has progressed, not how well a population is served by education.

Different educational stages

School provision cannot settle a language's university, research, practical-learning or accessibility needs.

Related communities

A need identified in one community prompts investigation of comparable communities. It does not justify copying their scores or assuming identical needs.

Research coverage is not educational coverage

The bounded intervention calculation currently contains package routes for 25 of the 100 forecast profiles, across 42 of 500 possible profile-stage cells. Those counts describe this calculation only: they omit separate shared-register analyses and do not measure the full research record, educational provision or need. The remaining cells are unfinished comparisons, not rejected populations. The wider retained register contains 8,618 language-level entries from Glottolog 5.3, including historical and bookkeeping entries labelled in the source. That register is an identity starting point, not an exhaustive inventory of every community, variety, script, diaspora or accessible format.

Twelve hypotheses to test

These expectations were written before the next comparison, after an earlier analysis. They are not an original preregistration. Evidence may weaken or overturn them.

H01

Large language communities with substantial stage-specific unmet need should repeatedly appear among high-reach candidates. Indonesian, Hindi, Bangla, Simplified Written Chinese and Urdu are strong initial expectations for investigation, not guaranteed winners.

Support or counterevidence: Compare affected learner counts and usable local provision at the same stage. High local provision or small additional benefit can weaken a particular package case; national wealth or local project completion cannot substitute for that evidence.

Unexpected absence: Trace the candidate through every inclusion, population, overlap and scoring step. Missing data cannot settle the comparison against it.

H02

Where a regional language supports schooling but advanced study moves to another language, bridge-to-tertiary resources may have especially high utility.

Support or counterevidence: Need evidence for the actual transition, learners' capabilities and subject gaps. NPTEL explicitly identifies this mechanism in India. Strong advanced local resources or comfortable academic-language access would weaken the predicted benefit.

Unexpected absence: Compare Tamil, Telugu, Marathi, Bangla, Odia and other relevant communities; do not infer one language's provision from another.

H03

A need mechanism found in one community should generate hypotheses for related educational systems and neighboring communities. Russian-language research or self-study gaps imply checking Ukrainian, Belarusian, Polish, Bulgarian and other relevant communities.

Support or counterevidence: Shared educational histories, academic registers or resource routes make an analogy worth testing. Conflict, displacement, different publication markets and different language policies may increase or decrease the gap. Genealogical relationship alone supplies neither an equal score nor a population estimate.

Unexpected absence: If Russian is present and Ukrainian absent, investigate source coverage, population definitions and implementation before interpreting the result. Repeat this test globally.

H04

Large national or census language categories may conceal distinct communities with different access needs.

Support or counterevidence: Compare the category definition with exact mother-tongue rows, scripts, schooling and bilingualism. Examples include Hindi-group varieties and Bengali categories containing Sylheti and Chatgaya. Shared classification is not measured comprehension.

Unexpected absence: Retain the group observation and its possible constituent populations. Resolve overlap or use bounds; never silently discard a constituent or credit it with the whole group.

H05

Educational-resource shortages differ by stage. A language can have extensive school resources and substantial postgraduate or research gaps.

Support or counterevidence: Compare curriculum coverage and source availability separately for school, vocational, university and research use. Japanese, Chinese, Russian, English, French and other established academic languages remain candidates for specific missing materials.

Unexpected absence: Investigate the proposed advanced package, rather than excluding the entire language as already served.

H06

A modest number of well-designed interlanguage editions could sometimes serve more learners per compute than separate editions. The advantage may be greater in formula-rich mathematics for readers with adequate prerequisites.

Support or counterevidence: Compare exact reader communities, scripts, mathematical background, prose burden, semantic-error patterns and standardized production cost. Evaluate definitions, negation and proof dependencies as well as formulas. Existing literature, source analysis and model/mechanical tests can support provisional scenarios without waiting for new participants.

Unexpected absence: Investigate bridge design and domain fit. Missing a comprehension estimate is not evidence of zero reach; family population is not evidence of full reach.

H07

Availability of some translated resources will often leave important curriculum and delivery gaps.

Support or counterevidence: Compare actual subjects, prerequisite sequences, exercises, answers, offline availability, accessibility and licensing. One catalogue is not a census of all provision; a listed file is not measured learner access.

Unexpected absence: Ask what is missing and what would add value, rather than treating either a catalogue entry or its absence as a final need estimate.

H08

Signed-language and accessibility interventions can create substantial access that written-language counts miss.

Support or counterevidence: Use sign-language-specific populations and educational evidence, not total deafness counts. Compare text, signed explanation, captions, MathML, audio and tactile routes by endpoint.

Unexpected absence: An entirely spoken/written result set requires an explicit modality audit. A missing written tokenizer is not a reason to remove signed learners.

H09

Offline, phone-usable, exercise-rich resources can benefit motivated learners without teachers, while classroom resources can help a different or overlapping audience.

Support or counterevidence: Estimate autonomous study, classroom use and research/reference use separately. Test file size, navigation, prerequisite support, answers and device requirements. Do not assume every person owns a smartphone or that a download guarantees learning.

Unexpected absence: Investigate independent-learning routes whenever a model implicitly requires teachers or continuous internet.

H10

Conflict, displacement, statelessness and cross-border language use can change both need and delivery routes without changing the language's registry homeland.

Support or counterevidence: Use dated displacement and education evidence, separate current location from language identity and avoid double-counting people across countries.

Unexpected absence: Census-country absence triggers a route search, not exclusion. Ukraine is one test; other displaced communities require the same treatment.

H11

The best source packages may differ materially across communities: foundational learning, science, computing, statistics, finance, management and advanced research may each dominate somewhere.

Support or counterevidence: Link package recommendations to local educational or practical needs and independent existing provision. OpenStax and Open Logic provide comparison baselines, not an exclusive subject universe.

Unexpected absence: A uniform proposal to translate the same book everywhere requires justification.

H12

Raw population, tokenization and missingness can each dominate a numerical model for the wrong reason.

Support or counterevidence: Run factor-removal, source-removal, cost-standardization, common-prior sensitivity and counterfactual missing-data tests. Keep gross, cached and fresh tokens distinct.

Unexpected absence: Investigate any ranking that is nearly a population table or that drops a community after evidence is removed.

The non-exhaustive candidate forecast

C001-C100 are identifiers, not ranks. Groups are research themes, not funding bands. All five stages remain eligible for every profile. The initial leads below are starting questions, not exclusions.

F = foundational/primary (including early learning); S = secondary; V = vocational/practical; U = university; R = postgraduate/research.

A. Broad-reach and major educational-language candidates

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C001	Indonesian Latin; Indonesia; L1 and additional-language users separately	F/S/V/U/R
C002	Hindi Devanagari; exact Hindi versus census-group mother tongues	F/S/V/U/R
C003	Bangla Bengali script; Bangladesh and India separately	F/S/V/U/R
C004	Standard Written Chinese, Simplified Mainland written-literacy profiles; not all Chinese varieties or all residents	S/V/U/R
C005	Urdu Perso-Arabic/Nastaliq; Pakistan and Indian communities separately	F/S/V/U/R
C006	Telugu Telugu script; regional school-to-tertiary routes	S/V/U/R
C007	Tamil Tamil script; India and Sri Lanka separately	S/V/U/R
C008	Marathi Devanagari; Maharashtra learning routes	S/V/U/R
C009	Vietnamese Vietnamese Latin orthography	S/V/U/R
C010	Hausa Latin first profile; Ajami separately; Nigeria and Niger separately	F/S/V/U
C011	Standard Kiswahili Latin; national standards and L1/L2 learning routes separately	F/S/V/U
C012	Modern Standard Arabic Learned written register; country, schooling and vernacular profiles separately	S/V/U/R
C013	Brazilian Portuguese Brazil; independent provision and disadvantaged learners	F/S/V/U/R
C014	Spanish, Latin American educational profiles Country-specific terminology and underserved routes, not a continent-wide count	F/S/V/U/R
C015	English, underserved learner profiles Territory and capability cells; targeted open/offline/accessible gaps	F/S/V/U/R
C016	French, underserved learner profiles African and other contexts separately; French ability cannot cover all residents	S/V/U/R
C017	Turkish Latin; Türkiye and diaspora separately	S/V/U/R
C018	Iranian Persian Perso-Arabic; distinguish Dari and Tajik	S/V/U/R

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C019	Thai Thai script; standard versus other Thai-area languages	S/V/U/R
C020	Burmese Myanmar script; distinguish other Myanmar languages	F/S/V/U
C021	Filipino Latin standard; Tagalog and other Philippine language access separately	S/V/U/R
C022	Standard Malay, Malaysia Latin profile; Jawi and other regional standards separately	S/V/U/R
C023	Gujarati Gujarati script	S/V/U/R
C024	Kannada Kannada script	S/V/U/R
C025	Odia Odia script; standard versus census-group scope	S/V/U/R
C026	Malayalam Malayalam script	S/V/U/R
C027	Eastern Punjabi Gurmukhi; do not substitute Western Punjabi/Shahmukhi counts	F/S/V/U
C028	Russian Cyrillic; resident and cross-border academic users separately	S/V/U/R
C029	Japanese Japanese writing; targeted subject and advanced-source gaps	U/R
C030	Korean, South Korean standard Hangul; North Korean and diaspora profiles separately	U/R

B. Regional and language-transition candidates

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C031	Amharic Ethiopic; do not substitute Ethiopia's total population	F/S/V/U
C032	West-Central Oromo educational profile Latin; other Oromo varieties separately	F/S/V/U
C033	Somali Latin standard; regional and displaced learners separately	F/S/V/U
C034	Yoruba Standard Latin with full marks; national and diaspora routes separately	F/S/V/U
C035	Igbo Standard Latin with tone/orthographic choices specified	F/S/V/U
C036	Kinyarwanda Latin; independent local supply audit	F/S/V/U
C037	Kirundi Latin; do not copy Kinyarwanda provision or reach	F/S/V/U
C038	Lingala School/written and urban spoken registers separately	F/S/V/U
C039	Central Kanuri Latin and Ajami routes separately	F/S/V
C040	Wolof Latin first profile; Wolofal and Garay separately	F/S/V/U
C041	Bambara Latin and N'Ko routes separately	F/S/V/U
C042	Pulaar Latin and Adlam profiles; Senegal/Mauritania and other contexts separately	F/S/V
C043	Nigerian Fulfulde Exact regional profile; no whole-Fulani population assignment	F/S/V
C044	Zulu Latin; school and higher-education transition	F/S/V/U
C045	Northern Sotho / Sepedi standard Latin; census and standard identity boundaries retained	F/S/V/U
C046	Haitian Creole Standard Haitian orthography; Haiti and displaced communities separately	F/S/V/U
C047	Nepali Devanagari; native and additional-language profiles separately	F/S/V/U
C048	Sinhala Sinhala script; specific Sri Lankan educational routes	S/V/U/R

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C049	Northern Pashto Perso-Arabic; Afghanistan/Pakistan and Southern Pashto separately	F/S/V/U
C050	Dari Perso-Arabic; Afghanistan and displaced learners separately	F/S/V/U
C051	Sindhi, Pakistan standard Perso-Arabic; Indian/Devanagari profiles separately	F/S/V/U
C052	Saraiki Perso-Arabic; no automatic Urdu or Punjabi coverage	F/S/V
C053	Bhojpuri Devanagari profile; Indian and Nepalese populations separately	F/S/V
C054	Maithili Devanagari profile; Tirhuta and territory distinctions retained	F/S/V/U
C055	Chhattisgarhi Devanagari; distinguish Hindi census grouping from comprehension	F/S/V
C056	Haryanvi Devanagari; actual Hindi overlap to estimate	F/S/V
C057	Santali Ol Chiki first profile; other script-literacy routes separately	F/S/V
C058	Sylheti Bengali-script and Sylheti-Nagri profiles separately	F/S/V
C059	Chittagonian Spoken and written-register design separately; no automatic Bangla coverage	F/S/V
C060	Javanese Latin first profile; registers and Indonesian overlap separately	F/S/V/U
C061	Sundanese Latin first profile; Indigenous-script route separately	F/S/V
C062	Cebuano Latin; Filipino/English proficiency measured separately	F/S/V/U
C063	Central Khmer Khmer script; Cambodia and other communities separately	F/S/V/U
C064	Lao Lao script; no automatic Thai coverage	F/S/V/U
C065	Northern Uzbek / Uzbekistan standard Latin and Cyrillic profiles; Southern Uzbek separately	S/V/U
C066	Kazakh Cyrillic first profile; actual script cohorts separately	S/V/U/R
C067	Tajik Cyrillic; do not assign Persian/Dari literacy automatically	S/V/U

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C068	Uyghur Perso-Arabic first profile; diaspora script routes separately	F/S/V/U
C069	Central Kurdish / Sorani Perso-Arabic; country and educational systems separately	F/S/V/U
C070	Northern Kurdish / Kurmanji Latin first profile; other scripts and regional varieties separately	F/S/V/U

C. Comparators, research access and regional-depth candidates

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C071	Ukrainian Cyrillic; residents and displaced learners separately	F/S/V/U/R
C072	Polish Latin; advanced gaps and refugee-learning routes separately	S/V/U/R
C073	Romanian Latin; Moldova and Romania separately	S/V/U/R
C074	Bulgarian Cyrillic; no Russian or Interslavic coverage assumption	S/V/U/R
C075	Belarusian Cyrillic first profile; actual preferred/usable educational language	S/V/U/R
C076	Serbian standard Cyrillic and Latin; other BCMS standards separately	S/V/U/R
C077	Tigrinya Ethiopic; Ethiopia, Eritrea and displaced learners separately	F/S/V/U
C078	Kabyle Latin first profile; Arabic/French overlap separately	F/S/V/U
C079	Central Atlas Tamazight Exact variety; Tifinagh/Latin routes and Moroccan standard separately	F/S/V
C080	Tachelhit Exact variety; Tifinagh/Latin/Arabic literacy separately	F/S/V
C081	Ayacucho Quechua Specified regional educational orthography	F/S/V
C082	Cusco Quechua Specified regional educational orthography; not all Quechua	F/S/V
C083	Paraguayan Guarani Standard orthography; bilingual and predominantly Guarani profiles	F/S/V/U
C084	Central Nahuatl Exact variety and local orthographic route; not all Nahuatl	F/S/V
C085	K'iche' Standard plus regional comprehension profiles	F/S/V
C086	Tok Pisin Standard written register; L1/L2 and schooling routes separately	F/S/V/U
C087	Bislama Standard written register; not all Vanuatu language users	F/S/V
C088	Kalaallisut / West Greenlandic Greenlandic standard; other Inuit languages separately	S/V/U/R

D. Signed-language hypotheses

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C089	American Sign Language US and other actual communities separately; not universal signing	F/S/V/U/R
C090	Brazilian Sign Language / Libras Regional signing variation and educational stages	F/S/V/U
C091	Indian Sign Language Regional variation and actual sign-user population; no national deafness substitution	F/S/V/U
C092	Kenyan Sign Language Actual Kenyan signing communities, not Kiswahili coverage	F/S/V/U
C093	South African Sign Language Regional variation; English/Zulu written access separately	F/S/V/U

E. Shared-register and interlanguage hypotheses

ID (not rank)	Edition/profile and distinctions to preserve	Initial leads
C094	Interslavic mathematical register Parallel Latin/Cyrillic; East/West/South Slavic reader profiles separately	U/R
C095	Hindi-Urdu shared mathematical register Devanagari/Nastaliq; register and literacy-specific reach	S/V/U
C096	Malay-Indonesian shared technical profile Exact national terminology choices; no wholesale Austronesian coverage	S/V/U
C097	Persian-Dari-Tajik shared technical profile Perso-Arabic/Cyrillic; lexical and schooling differences explicit	S/V/U/R
C098	Oghuz-focused Inter-Turkic design hypothesis Turkish/Azerbaijani/Turkmen profiles; not all Turkic languages	U/R
C099	N'Ko literary Manding educational profile Exact Manding communities and N'Ko literacy; no family-wide reader count	F/S/V/U
C100	Romance mathematical intercomprehension profile Exact Portuguese/Spanish/French/Italian/Romanian routes; natural-text and constructed alternatives compared	U/R

Candidates beyond the forecast

This shortlist is a prediction to challenge, not the universe from which the result must be selected. Chinese Traditional profiles, Western Punjabi, Southern Pashto, Egyptian and other spoken Arabic varieties, Xhosa, Chichewa, Luganda, Akan standards, Fijian, Samoan, Tetum, Mongolian, Armenian, Georgian, Czech, Slovak, Croatian, Bosnian, German, Italian, Ukrainian and other signed languages, Romani varieties, additional Quechua/Nahuatl/Mayan/Inuit varieties and many others can enter the final portfolio if supported. They are not assigned a score of zero or a rank below 100 by being outside this provisional table.

The proposed Inter-Germanic concept remains an explicit challenger. Investigate actual incremental access for Low German, Plautdietsch, Pennsylvania Dutch, Hutterisch, Frisian, Afrikaans, Yiddish, Scots, Faroese and other relevant profiles. Compare West/North or narrower designs against one pan-Germanic design. Do not use German/Dutch/Nordic English proficiency as a substitute for those communities' capabilities.

No group's need is removed by a bridge proposal. Native-language and bridge editions are competing or complementary interventions on overlapping populations. Report standalone benefit separately from portfolio benefit; never sum all potential audiences without overlap adjustment.

What happens next

- Expand comparable population, stage, subject, supply and delivery evidence beyond the initial forecast.

- Test useful full courses and smaller additions on the same basis, including OpenStax, Open Logic, other open resources and advanced research materials.

- Compare native-language, interlanguage, signing and accessible-format options without assuming complete overlap or zero comprehension.

- Investigate unexpected absences and possible high-value candidates outside the forecast; test the responsible rule across the wider universe.

- Develop evidence-supported recommendations with uncertainty, rather than converting research order or missing measurements into ranks.

Initial source basis and limits

The World Bank's 2021 *Loud and Clear* identifies language-of-instruction mismatch as a barrier and reports nearly 37% of students in low- and middle-income countries taught in a language they do not understand. This supports investigating H02; it is not a rate to multiply by every language's population. World Bank.

UNESCO's multilingual-education overview describes a global access problem, including migration and language diversity. Its 40% population figure and the World Bank student figure have different denominators. Neither supplies a language-by-language ranking. UNESCO.

India's 2011 C16 distinguishes census language groups from named mother tongues; C17 reports up to two additional languages. These provide population and reporting evidence, not direct measures of comfortable academic access. C16, C17.

NPTEL describes difficulties moving from regional-language schooling to English technical education and provides translations in 11 languages. This is direct institutional support for H02 and evidence of existing supply relevant to H07, not proof that all needed material is available. NPTEL.

ONS's detailed language classification groups Bengali with Sylheti and Chatgaya. This is a concrete example of why a census category cannot by itself establish a single edition's coverage. ONS TS024.

The Unicode Standard describes N'Ko as a literary Manding register and script. This makes a shared written-profile hypothesis concrete, but a Manding speaker count is not a count of N'Ko-literate learners. Unicode 17.0, section 19.4.

These sources motivate mechanisms. They do not validate every candidate in the table or establish the eventual order. Country-, language-, stage- and package-specific investigation remains required.

Snapshot and authorship

AI-assisted research notebook published by the research account KokunoYumeto. The source bundle contains the frozen protocol, candidate register, coverage table and wider identity register. No transcript, credential or private working-state file is included.

This is ongoing, non-exhaustive research. It cannot be claimed as the final product.